

Cross-Sectional Factor Momentum: Evidence from Multiple Formation Periods

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Abstract

We revisit cross-sectional factor momentum by exploring a range of formation periods. Recent- and intermediate-past horizons, jointly and severally, not only produce statistically and economically significant factor momentum, remaining robust to established factors, but also capture its distinct aspects, with neither spanning the other. Subsample analyses show that the recent-past factor momentum premium persists post-publication, suggesting it is unlikely driven by data mining, while the intermediate-past premium varies with the publication and significance of factors.

Keywords: Cross-sectional factor momentum; Formation periods; Publication effect; Term structure

JEL Classification: G12; G14

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1 Introduction

Momentum is a pervasive anomaly in financial markets, attracting significant attention from both academics and practitioners. While early research focused on momentum in stock returns, recent studies have extended the concept to factor returns. [Ehsani and Linnainmaa \(2022\)](#) investigate time-series (TS) factor momentum, where factors' past performance predicts their future returns. They find that factor momentum subsumes stock momentum and argue that momentum is not a distinct risk factor but rather a combination of autocorrelations present in other factors.

[Arnott et al. \(2023\)](#) examine not only TS but also cross-sectional (CS) factor momentum. They highlight that, while TS strategies are net-long investments, CS strategies are zero-net investments.¹ After controlling the former net-long positions, the authors show that both strategies yield comparable performance. Their factor momentum considers 1-, 6-, and 12-month formation periods as alternatives for robustness checks. However, the distinct nature and interplay among these periods remain relatively underexplored. Meanwhile, [Novy-Marx \(2012\)](#) revisits stock momentum with recent- and intermediate-past horizons, using (2-6)- and (7-12)-month periods, respectively, and finds the latter exhibits stronger predictability.²

Motivated by these findings, we reexamine CS factor momentum across 1- to 60-month formation periods. By exploring strategies over various horizons, we seek to deepen our understanding of momentum dynamics and assess whether varying formation periods offer overlapping or distinct predictability. Revealing the dual nature of recent- and intermediate-past factor momentum, we find the strongest performance with the 1-month strategy, followed by the (2-6)- and (7-12)-month strategies. This finding aligns with both the short-term characteristics of factor momentum ([Arnott et al., 2023](#)) and the significance of intermediate-past momentum ([Novy-Marx, 2012](#)). In detail, [Novy-Marx \(2012\)](#) highlights the importance of the intermediate-past horizon for stock-level CS momentum; in contrast, we underscore the importance of both recent- and intermediate-past horizons for factor-level CS momentum.

For robustness checks, we conduct extensive subsample analyses, focusing on the pre- and post-publication periods of the original studies documenting the factors, inspired by [McLean and Pontiff \(2016\)](#). If factors genuinely proxy for risk, as argued by [Fama and French \(2015\)](#), we would expect the post-publication premium to remain significant. We show the recent-past factor momentum premium persists post-publication, suggesting it is less likely driven by data mining or statistical biases, while the intermediate-past factor momentum premium varies depending on the publication and significance of factors. Furthermore, our double-sort analyses of recent- and intermediate-past horizons indicate that they capture distinct dimensions of this anomaly, underscoring the importance of considering multiple formation periods in factor momentum research.

¹See [Goyal and Jegadeesh \(2018\)](#) for a comparison of TS and CS momentum.

²See [Sharifkhani and Simutin \(2021\)](#) for a recent explanation of this effect through industry trade networks.

Our study makes novel contributions to the emerging literature on factor momentum.³ First, we offer systematic analyses of CS factor momentum across broad formation periods, building on existing evidence (Gupta and Kelly, 2019; Grobys et al., 2022; Arnott et al., 2023), providing new insights into its time-varying nature. Second, we conduct rigorous robustness checks pre- and post-publication, addressing concerns about the persistence of predictability (McLean and Pontiff, 2016; Liu et al., 2023). While our findings support the robustness of recent-past momentum, we remain cautious in interpreting them as evidence of risk compensation. Third, our double-sort analyses uncover the non-overlapping aspects of recent- and intermediate-past momentum (Novy-Marx, 2012; Goyal and Wahal, 2015; Trigilia and Wang, 2019), reinforcing the need to incorporate various formation periods.

The remainder of the paper proceeds as follows: Section 2 describes data and methodology. Section 3 presents main results and robustness checks. Section 4 concludes.

2 Data

We obtain U.S. value-weighted monthly factors for 153 characteristics, spanning January 1926 to December 2023, from Jensen et al.’s (2023) website, as basis assets for CS factor momentum, following literature (Arnott et al., 2023; Cakici et al., 2025). Each factor represents a long-short strategy that buys (sells) the value-weighted top (bottom) tercile portfolio sorted by a given characteristic. Figure A1 in the appendix shows the average returns and standard deviations of these factors.

In constructing the factor momentum, we sort these factors into equal-weighted tercile portfolios by the factors’ 1-month lagged returns. We also sort the tercile portfolios by previous (2-6)-, (7-12)-, and (13-60)-month returns—which do not overlap with the aforementioned 1-month returns—after considering prior 1 to 60 months, to investigate the short- and long-term dynamics of the factor momentum.⁴ Then, we form the winner-minus-loser (WML) factor momentum strategy by buying and selling the top- and bottom-tercile portfolios, respectively, and holding that position for one month and rebalancing monthly.

³See also, among others, Yan and Yu (2023), Fieberg et al. (2024), Käfer et al. (2024), Cakici et al. (2025) for recent discussions on factor momentum.

⁴In other words, in terms of Novy-Marx’s (2012) n-m strategy, which refers to the strategy based on portfolios sorted on returns from n to m months prior to their formation, we consider 1-1, 2-6, 7-12, and 13-60 strategies.

3 Results

3.1 Fama–MacBeth Regressions

We examine the predictability of past factor returns across different horizons using Fama–MacBeth regressions. Table 1 presents their results, with Panel A including all factors and Panel B focusing on those significant in [Jensen et al. \(2023\)](#). The 1-month lagged return has the strongest predictability in both Panels A and B, with t-statistics of 4.73 and 4.80, respectively. The (2-6)- and (7-12)-month returns also show significant predictability, with t-statistics ranging from 2.46 to 2.90. However, the (13-60)-month return is significant in neither panel. When all regressors are included, the 1-, (2-6)-, and (7-12)-month returns remains significant in both panels, suggesting their joint predictability.

3.2 Single-Month Strategy Performance

To assess predictability at each of the 1- to 18-month horizons, we reexamine single-month momentum strategies. Figure 1 presents their results. The 1-month strategy generates the highest average return of 6.28%, consistent with our Fama–MacBeth results, and has a Sharpe ratio of 0.41, with a relatively low standard deviation of 15.44%. The 2-, 11-, and 12-month strategies also show significant performance, generating average returns of 3.17%, 3.63%, and 2.14% with Sharpe ratios of 0.22, 0.29, and 0.18, respectively. These findings reinforce the dual nature of recent- and intermediate-past factor momentum. Interestingly, the 17-month strategy also shows significant performance, with its CAPM and Fama–French alphas in Figure A2 in the appendix further supporting its robustness.

Overall, our analyses support CS factor momentum, particularly over recent- and intermediate-past horizons. The 1-, 2-, 11-, and 12-month strategies emerge as the most profitable, while the 17-month strategy exhibits significance as well. These findings highlight the uniqueness of each horizon, underscoring the need for multiple formation periods.

3.3 Subsample Analyses

Recently, [Liu et al. \(2023\)](#) document that the statistical significance of basis factors affects the performance of factor momentum. Since the publication status of factors influences both their statistical and economic significance ([McLean and Pontiff, 2016](#)), the factor momentum effect may also vary with publication status. To address this concern and perform robustness checks, we conduct subsample analyses using the publication years of the cited studies documenting the factors ([McLean and Pontiff, 2016](#)). We divide our sample into four subsamples: pre-publication (Pre-Cite), post-publication (Post-Cite), pre-publication with significant

factors (Pre-Cite-Sig), and post-publication with significant factors (Post-Cite-Sig). Pre-Cite (Post-Cite) includes each factor until (after) the publication year of its original study, excluding factors without citation information. The significant factor subsamples include those significant at the 10% level post-publication.⁵

Table 2 repeats Model 5 in Table 1 with these subsamples and presents their results. The 1-month return remains the most significant across all subsamples, with t-statistics ranging from 2.74 to 3.94. The (2-6)- and (7-12)-month returns are also significant in Pre-Cite, with t-statistics of 3.91 and 1.83, respectively. The former remains significant in Pre-Cite-Sig, while the latter does not. Both lose predictability post-publication. The (13-60)-month return is insignificant across all subsamples except Post-Cite-Sig. Adjusted r-squareds, ranging from 0.2974 to 0.4487, indicate that the joint predictability of lagged returns persists.

Overall, our analyses support the predictability of recent- and intermediate-past CS factor momentum and their dual nature. While the 1-month horizon remains the strongest, the (2-6)- and (7-12)-month horizons also make significant contributions. The influence of factors' publication and significance is marginal at the recent-past horizon but substantial at the intermediate-past horizon.

3.4 Double-Sorted Portfolios

Building on the Fama–MacBeth and single-month results above that suggest the distinct contributions of recent- and intermediate-past horizons around months 1 and 11, respectively, we double-sort portfolios by the 1- (r_1) and (7-12)-month ($r_{7,12}$) lagged returns, similar to Novy-Marx (2012), to examine the interplay between these horizons in greater detail.⁶ Table 3 presents their results. Panels A and B show that the WMLs single-sorted by r_1 and $r_{7,12}$ generate significant average returns of 6.28% and 2.97%, respectively. Panel C exhibits the WMLs independently double-sorted by r_1 and $r_{7,12}$. We find that these WMLs generate significant returns in all terciles of the other return except r_1 Loser, suggesting that r_1 and $r_{7,12}$ represent distinct dimensions of momentum. In the appendix, Panels A and B of Table A3 present the WMLs dependently double-sorted by r_1 and $r_{7,12}$ first, respectively, and then the other return. These results align with those in Panel C. Consistent with these results, Table A1 shows the CAPM and Fama–French alphas of each return in Table 3, and Table A2 shows the spanning alphas of the two single-sorted WMLs above.

Overall, our results indicate that recent- and intermediate-past factor momentum capture non-overlapping aspects, with neither subsuming the other.

⁵For example, Jensen et al. (2023) cite Lakonishok et al. (1994) for the free cash flow-to-price factor. It is split into 530 Pre-Cite (November 1950–December 1994) and 348 Post-Cite (January 1995–December 2023) observations. Its Post-Cite t-statistic of 3.44 further classifies it in Pre-Cite-Sig and Post-Cite-Sig.

⁶The results remain robust when $r_{1,2}$ replaces r_1 .

4 Conclusion

We explore CS factor momentum using broad formation periods from 1 to 60 months. The 1-, (2-6)-, and (7-12)-month strategies yield significant and robust returns, in that order. Our double-sort analyses of the 1- and (7-12)-month strategies reveal their distinct nature, emphasizing the need to consider multiple formation periods in factor momentum studies. As neither fully spans the other, these results are somewhat at odds with [Novy-Marx’s \(2012\)](#), which show the latter dominating the former at the stock level. The results in the appendix further reveal that, while these strategies share commonalities with standard momentum, they also feature distinct loadings on the Fama–French factors.

Our subsample analyses show the persistence of the 1-month factor momentum premium, suggesting it is unlikely driven by data mining. We recognize, however, that our findings may be influenced by investor biases and transaction costs.⁷ Thus, we interpret them with caution and do not claim this premium as definitive risk compensation. The analyses also show that the (2-6)- and (7-12)-month premia vary with factors’ publication and significance, reinforcing the distinct aspects of recent- and intermediate-past factor momentum.

Our study contributes by analyzing CS factor momentum across extensive formation periods, conducting robust post-publication checks, and highlighting the joint characteristics of recent- and intermediate-past factor momentum. Future research should investigate their economic drivers and implications for asset pricing theories and investment practices.

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⁷Although not tabulated, the results of transaction cost-adjusted factor momentum ([Gupta and Kelly, 2019](#)) and bimonthly-rebalanced factor momentum ([Moorman, 2014](#)) suggest that the findings remain reasonably robust net of transaction costs.

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Table 1. Fama–MacBeth Regressions

This table shows results from Fama–MacBeth regressions of [Jensen et al.’s \(2023\)](#) factors on their past performance. Past 1-, (2-6)-, (7-12)-, and (13-60)-month returns accumulate factor plus risk-free rates. Panel A includes all monthly factors, and Panel B exclude those insignificant in [Jensen et al. \(2023\)](#) or without significance information. Regressions span February 1926 to December 2023, exclude months with fewer than 10 observations, winsorize each month at the 0.5th and 99.5th percentiles, and estimate Newey–West standard errors with 12 lags. t-statistics are in parentheses, and ***, **, and * mean significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Results with all factors

Variable	Coefficient and (test-statistic) from regressions of the form $r_{it} = \beta' \mathbf{x}_{it} + \varepsilon_{it}$				
	Model 1	Model 2	Model 3	Model 4	Model 5
Intercept	0.0011*** (4.58)	0.0012*** (4.44)	0.0014*** (4.61)	0.0013*** (2.71)	0.0003 (1.07)
Return 1	0.0988*** (4.73)				0.0749*** (4.20)
Return 2-6		0.0214*** (2.86)			0.0202*** (4.33)
Return 7-12			0.0160** (2.57)		0.0099** (2.18)
Return 13-60				0.0020 (1.28)	0.0015 (1.58)
Adjusted r-squared	0.2096	0.1847	0.1616	0.1041	0.4342
Number of observations	1,169	1,164	1,158	1,110	1,110

Panel B. Results with factors significant in [Jensen et al. \(2023\)](#)

Variable	Coefficient and (test-statistic) from regressions of the form $r_{it} = \beta' \mathbf{x}_{it} + \varepsilon_{it}$				
	Model 1	Model 2	Model 3	Model 4	Model 5
Intercept	0.0013*** (4.86)	0.0013*** (4.69)	0.0016*** (4.71)	0.0016*** (2.95)	0.0004 (1.37)
Return 1	0.1015*** (4.80)				0.0768*** (4.23)
Return 2-6		0.0220*** (2.90)			0.0206*** (4.45)
Return 7-12			0.0155** (2.46)		0.0095** (2.11)
Return 13-60				0.0019 (1.14)	0.0014 (1.42)
Adjusted r-squared	0.2116	0.1838	0.1606	0.1059	0.4357
Number of observations	1,169	1,164	1,158	1,110	1,110

Table 2. Subsample Fama–MacBeth Regressions

This table repeats Model 5 in Table 1 with four subsamples: Pre-Cite, Post-Cite, Pre-Cite-Sig, and Post-Cite-Sig. The first (second) column includes factors until (after) their original citation years, while the third (fourth) column includes these factors significant post-citation. Refer to Table 1 for other regression details.

Variable	Coefficient and (test-statistic) from regressions of the form $r_{it} = \beta' \mathbf{x}_{it} + \varepsilon_{it}$			
	Pre-Cite	Post-Cite	Pre-Cite-Sig	Post-Cite-Sig
Intercept	0.0004 (1.21)	-0.0003 (-0.44)	0.0010 (1.48)	0.0004 (1.09)
Return 1	0.0745*** (3.87)	0.0918*** (3.94)	0.0568*** (2.74)	0.1074*** (3.40)
Return 2-6	0.0195*** (3.91)	0.0124 (1.65)	0.0120* (1.88)	0.0008 (0.07)
Return 7-12	0.0089* (1.83)	0.0146 (1.59)	0.0094 (1.45)	0.0083 (0.84)
Return 13-60	0.0008 (0.79)	0.0018 (0.90)	-0.0008 (-0.53)	0.0043* (1.95)
Adjusted r-squared	0.4284	0.4487	0.3728	0.2974
Number of observations	1,026	420	638	216

Table 3. Performance of Portfolios Sorted by r_1 and $r_{7,12}$

This table reports the annualized average monthly returns of equal-weighted tercile factor momentum portfolios sorted by factors' past 1- (r_1) and (7-12)-month ($r_{7,12}$) returns. Panel A (B) sorts tercile portfolios by r_1 ($r_{7,12}$). Panel C sorts 3-by-3 portfolios independently by r_1 and $r_{7,12}$; refer to Table A3 in the appendix for dependently double-sorted portfolios. [Jensen et al.'s \(2023\)](#) factors span January 1926 to December 2023; accordingly, portfolios sorted by r_1 and $r_{7,12}$ begin in February 1926 and January 1927, respectively. t-statistics are in parentheses, and ***, **, and * mean significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Single-sort by r_1

Sorted by	Loser	Mid	Winner	WML
r_1	-1.51 (-1.63)	2.14*** (4.49)	4.77*** (5.33)	6.28*** (4.02)

Panel B. Single-sort by $r_{7,12}$

Sorted by	Loser	Mid	Winner	WML
$r_{7,12}$	0.40 (0.50)	1.66*** (3.18)	3.36*** (3.59)	2.97** (2.06)

Panel C. Independent double-sort by r_1 and $r_{7,12}$

Sorted Independently by	$r_{7,12}$ Loser	$r_{7,12}$ Mid	$r_{7,12}$ Winner	$r_{7,12}$ WML
r_1 Loser	-1.16 (-1.15)	-1.26 (-1.44)	-0.01 (-0.01)	0.78 (0.64)
r_1 Mid	0.81 (1.03)	1.65*** (2.85)	3.44*** (3.83)	2.55** (2.01)
r_1 Winner	3.42*** (3.19)	4.29*** (5.10)	6.36*** (5.92)	2.77** (2.28)
r_1 WML	4.65*** (2.82)	5.56*** (4.18)	5.48*** (4.33)	

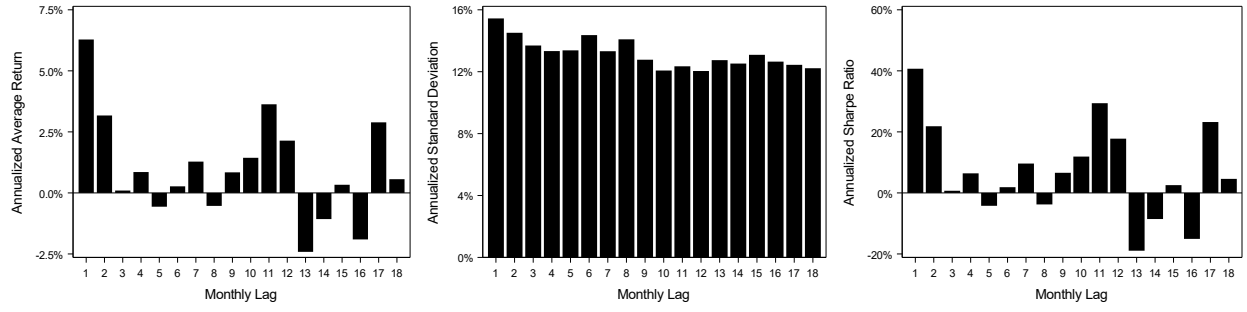


Figure 1. Single-Month Strategy Performance

This figure shows the annualized average monthly returns (left), standard deviations (center), and Sharpe ratios (right) for winner-minus-loser strategies based on single-month performance. Winners and losers are the equal-weighted top- and bottom-tercile portfolios, respectively, of single-month performance, beginning from lag months before their formation. The 1-, 2-, and 11-month strategies show average returns of 6.28% (t-statistic=4.02), 3.17% (2.16), and 3.63% (2.89), with Sharpe ratios of 0.41, 0.22, and 0.29, respectively. [Jensen et al.'s \(2023\)](#) factors span January 1926 to December 2023.